DETECTING BARS IN GALAXY PAIRS AND COMPARING THEIR PROPERTIES WITH ISOLATED GALAXIES

Internship Project Report

Submitted by

VIKRANT GUPTA

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IISER Pune



Under the guidance of

Dr. MOUSUMI DAS

Professor

Indian Institute of Astrophysics, Bangalore

Mentor

Ms. SAILI KESHRI

PhD Research Scholar

Indian Institute of Astrophysics, Bangalore



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Abstract

In this project, we study how interactions between galaxies affect the shape and growth of stellar bars. While galaxies usually change very slowly on their own, the strong tidal forces from a close neighbor can speed up these structural changes. We analyze a sample of interacting barred galaxies selected from the Arp-Madore Catalog. We use near-infrared K_s-band images from VISTA to bypass dust interference, we clean the data by masking out foreground stars and removing the sky background. We then use `Photutils` to fit ellipses along the galaxy isophotes.

After the isophotal ellipse fitting, we extracted structural parameters like, bar length, bar position angle, galaxy position angle and galaxy inclination, this is then used to deproject the observed bar length values into true, face-on physical bar lengths. We also used the total integrated K_s-band fluxes within this boundary, combined with redshifts, to calculate absolute magnitudes and estimate the total stellar mass of each galaxy. We filtered our dataset to keep only galaxies with intermediate inclinations ($20^\circ \leq i \leq 60^\circ$) to ensure reliable measurements.

Finally, we looked at how the normalized bar length changes depending on the interacting galaxy mass ratio (q). Our results reveal an environmental split: dual-barred pairs (where both interacting galaxies are barred) only appear in major mergers ($q \geq 0.3$), while minor mergers ($q < 0.3$) completely lack them. Furthermore, we found two completely opposite trends. Single-barred galaxies (where only one galaxy is barred in the interacting pair) show a positive correlation, meaning stronger tidal forces lead to longer bars. In contrast, dual-barred systems show a negative correlation where bars get shorter in nearly equal-mass mergers.

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1 Introduction

1.1 Morphology of Galaxies

The structural classification of galaxies serves as the foundational framework for understanding their formation pathways, stellar populations, and dynamical evolution. The most enduring systematic convention is the Hubble sequence, introduced by Edwin Hubble. This geometric scheme organizes galaxies based on their visual appearance in optical imaging into a progression popularly referred to as the Hubble tuning fork.

The sequence begins with early-type galaxies, which are dominated by smooth, pressure-supported stellar distributions with negligible cold gas reservoirs and ongoing star formation. These are divided into:

- **Elliptical Galaxies (E0–E7):** Visually featureless systems characterized by triaxial, ellipsoidal geometries. They are classified analytically by their projected ellipticity, determined by the ratio of their minor axis (b) to major axis (a) via the relation En = $10(1 - b/a)$. A round system is designated as E0, while the most flattened elongated systems reach up to E7. [2]
- **Lenticular Galaxies (S0 and SB0):** These are transition galaxies that harbor a prominent, concentrated central bulge surrounded by a distinct, rotationally supported stellar disk. These galaxies lack well-defined spiral arms or large quantities of dust and interstellar gas, representing a structural bridge between ellipticals and spirals.

Branching outward from the lenticular junction, the sequence splits into two parallel tracks of late-type, rotationally supported disk galaxies:

- **Spiral Galaxies (Sa–Sc):** Systems possessing a central bulge and a distinct disk where bright spiral arms wind outward. The progression from Sa to Sc tracks a systematic decrease in the relative size and luminosity of the central bulge, an increasing open geometry (unwinding) of the spiral arms, and an increase in atomic gas mass fractions and active star formation.
- **Barred Spiral Galaxies (SBa–SBc):** Parallel to the spirals, these systems feature a distinct, linear stellar structure, a bar, that bisects the central nucleus before terminating at the starting points of the spiral arms. Similar to the unbarred track, the sub-classifications (SBa, SBb, SBc) track the changing prominence of the central bulge and the tightness of the outer arm structures.

At the apex of the tuning fork sit **Irregular Galaxies (Irr)**. These systems lack any rotational symmetry, ordered spiral structures, or prominent central nuclei. Instead, their morphologies are chaotic and dominated by patchy, highly localized starburst regions, often driven by high gas fractions or external gravitational disruptions. [1]

1.2 Interactions of Galaxies

While the classic Hubble sequence describes galaxies as isolated, static systems, galaxies in the real universe frequently interact with one another due to gravity. Galaxies are rarely isolated; they cluster together in groups and pairs where the distances between them are relatively small compared to their actual sizes. When two or more galaxies pass close to one another, their mutual gravitational pull triggers tidal forces that permanently reshape their structures.

During a close flyby or a merger event, the gravitational pull on the near side of a galaxy is significantly stronger than the pull on its far side. This gravitational gradient creates a stretching effect known as tidal stripping. As a result, stars, gas, and dark matter are ripped away from the main galactic disks and pulled into the intergalactic medium. This process produces large-scale structures like:

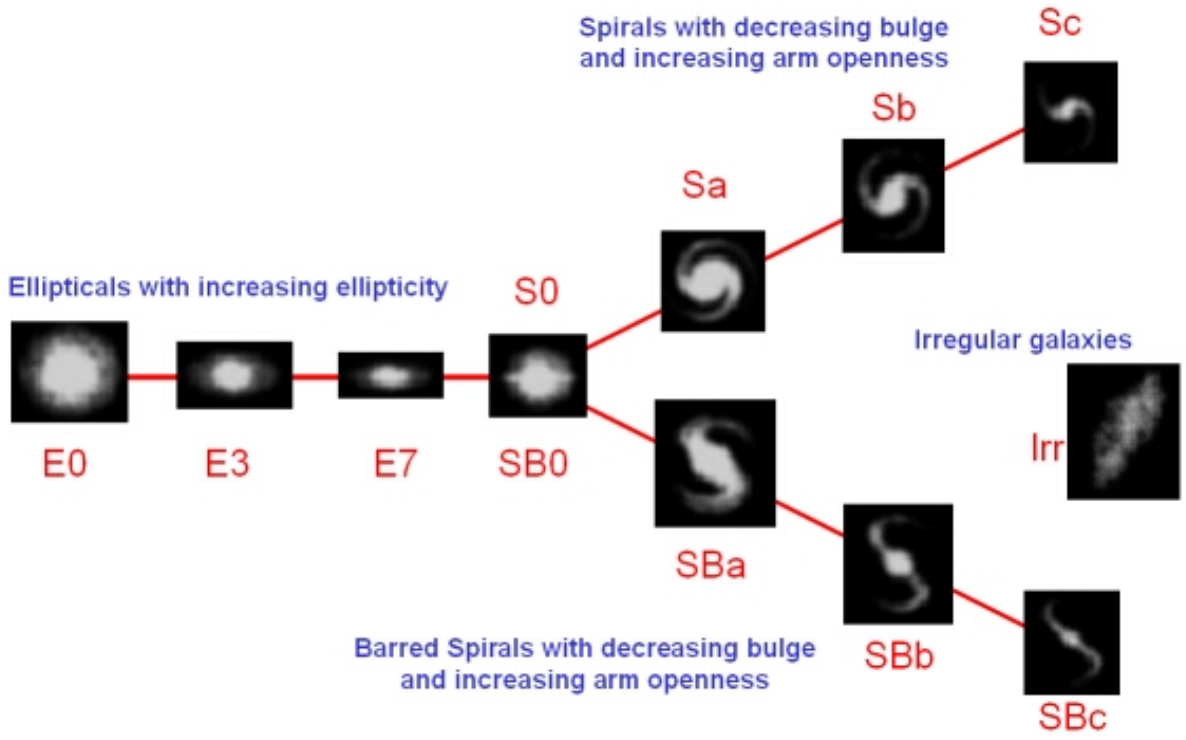


Figure 1: The Hubble tuning fork diagram. It transitions from elliptical galaxies (E0–E7) into lenticulars (S0/SB0), where it splits into normal spirals (Sa–Sc) and barred spirals (SBa–SBc) as central bulges decrease and arms loosen. Irregular galaxies (Irr) sit on the far right.
Credit: astronomy.swin.edu.au/cosmos

- **Tidal Tails:** Elongated streams of stars and gas ejected into the intergalactic medium due to gravitational forces, mapping out the trajectory of the encounter.
- **Bridges:** Connecting structures of stellar and gaseous matter that span the space between the two interacting nuclei, bridging the companion galaxies. [3]

Beyond changing the visual shape of a galaxy, gravitational interactions act as a massive catalyst for internal cosmic evolution. The strong tidal forces disrupt the stable, circular orbits of gas clouds within the galactic disk, removing their angular momentum. This causes vast amounts of cold gas to lose its orbital energy and plunge inward toward the central nuclear regions of the galaxy.

This rapid gas inflow has two major consequences. First, the sudden concentration of dense gas in the core triggers intense, widespread starburst activity, creating dense clusters of newborn stars. Second, if the gas reaches the absolute center, it can feed the supermassive black hole, turning the galaxy into an Active Galactic Nucleus (AGN). [4] [5]

1.3 Dynamics and Evolution of Galactic Bars

1.3.1 Physical Nature

A galactic bar is a non-axisymmetric density perturbation within the disk of a galaxy, rather than a rigid structural component. It represents a dynamic equilibrium sustained by the collective gravitational potential of billions of stars.

In an unbarred disk, stars generally follow near-circular orbits where the centripetal force is balanced by the radial gravitational force of the galaxy. A bar forms when these circular paths are perturbed and transition into elongated, eccentric orbits. When the major axes of



Figure 2: Examples of galaxy morphologies and tidal interactions. (Top) Ground-based view of the actively merging Antennae Galaxies (NGC 4038/4039), showing extended tidal tails. (Middle) HST ACS observation of the interacting Mice Galaxies (NGC 4676), displaying a tidal tail and a bridge. (Bottom) HST view of the prototypical grand-design barred spiral galaxy NGC 1300, featuring a distinct linear stellar bar with spiral arms originating from its ends. Credits: ESA/Hubble, R. Gendler; NASA, ESA, ACS Science Team, Hubble Heritage Team.

these elliptical orbits align in space, a region of localized high density is created. As stars pass through the long ends of their elliptical paths, they slow down due to conservation of angular momentum, spending more time there and physically reinforcing the visual and gravitational structure of the bar. [6]

1.3.2 Formation Mechanisms

The formation of a bar is fundamentally a problem of dynamical stability in rotationally supported systems.

In astrophysics, a disk is considered "kinematically cold" if the random, non-circular velocities (velocity dispersion) of its stars are low compared to their systematic rotational velocity. Cold disks are highly susceptible to gravitational instabilities; slight density ripples can grow exponentially because the stars lack sufficient random kinetic energy to resist grouping together.

Conversely, a "hot" disk possesses high velocity dispersion. The random kinetic motion acts as a stabilizing pressure, suppressing the growth of non-axisymmetric structures. Additionally, a massive, centrally concentrated spherical component, such as a dark matter halo or a stellar nuclear bulge, can stabilize the disk against bar formation by dominating the central gravitational potential. [7]

For a bar instability to successfully grow and propagate, the disk must possess a sufficiently high stellar surface density. High stellar density ensures strong self-gravitational attraction between stars within the disk plane. This localized gravitational coupling overcomes global shear forces, providing the necessary self-gravity to bind and sustain the elongated stellar orbits that constitute the bar.

1.3.3 Angular Momentum Exchange with the Dark Matter Halo

Once a bar is established, it interacts gravitationally with the surrounding dark matter halo. Because the bar is non-spherical, it exerts a continuous gravitational drag on the halo.

As the bar rotates through the halo, it creates a trailing gravitational wake in the dark matter distribution. This wake pulls backward on the bar, exerting a negative torque. As a result, the bar loses angular momentum and its rotation rate (pattern speed) slows down over time.

The efficiency of this braking depends heavily on the initial rotational state of the dark matter halo. If the halo has low initial angular momentum, it acts as an efficient sink for the bar's rotational energy, allowing the bar to grow longer and more massive. If the halo is already rotating rapidly in the same direction as the disk, the relative velocity between the bar and the halo is lower, reducing the frictional drag and limiting bar growth. [8]

1.3.4 Bar Dissolution

The presence of a bar fundamentally alters the long-term, secular evolution of a galaxy by acting as a mechanism for redistributing mass and angular momentum.

Unlike collisionless stars, interstellar gas clouds experience hydrodynamic dissipation and shocks when they encounter the non-axisymmetric gravitational potential of the bar. The bar exerts strong gravitational torques on the gas, stripping it of its orbital angular momentum. Deprived of the velocity needed to maintain its orbit, the gas loses energy and funnels inward toward the galactic center. [9]

This continuous accumulation of gas builds a highly dense central mass concentration. As this central mass grows, it modifies the central gravitational potential of the galaxy, making it increasingly spherical. Eventually, the gravitational pull of this central core becomes strong enough to disrupt the aligned, elliptical stellar orbits that support the bar. The orbits become

chaotic, the bar dissolves, and the system transitions into a centralized, spheroidal pseudobulge. [10]

1.4 Motivation

The structural evolution of disk galaxies is governed by both internal secular processes and external environmental influences. While internal mechanisms, such as the formation of stellar bars redistribute mass and angular momentum over long timescales, environmental mechanisms like tidal interactions and mergers alter galaxy morphologies on much shorter, dynamical timescales. Although single-barred systems in isolated environments have been extensively classified, the behavior, survival, and structural parameters of bars within actively interacting systems remain poorly constrained. This study is motivated by the need to observationally quantify how environmental stress influences the fundamental physical properties of galactic bars.

A primary motivation of this project is to determine how tidal forces and gravitational perturbations during galaxy interactions affect the physical dimensions of bars. By conducting an observational analysis to measure parameters like, bar lengths and galaxy mass ratios, this study seeks to establish empirical trends unique to interacting environments. Comparing these parameters directly against control samples of isolated single-barred systems allows us to isolate the specific effects of environmental factors from purely internal evolutionary pathways. Understanding whether tidal interactions systematically elongate, truncate, or destabilize these structures provides critical insights into the resilience of non-axisymmetric perturbations under external gravitational strain.

1.5 Data

1.5.1 DECaLS DR9 and DR10

The optical imaging data utilized in this study are sourced from the Dark Energy Camera Legacy Survey (DECaLS), incorporating both Data Release 9 (DR9) and Data Release 10 (DR10) footprints. The imaging is obtained using the Dark Energy Camera (DECam), a prime-focus optical imager mounted on the Víctor M. Blanco 4-meter Telescope at the Cerro Tololo Inter-American Observatory (CTIO) in Chile. DECam features a 520-megapixel focal plane array composed of 62 science CCDs, delivering a pixel scale of $0.262''$/pixel and a wide field of view spanning approximately $3\,\mathrm{deg}^2$.

Photometric observations are conducted across three primary optical bands: g, r, and z. These filters provide excellent depth limits, reaching 5σ point-source magnitudes of approximately 24.0, 23.4, and 22.5, respectively. The broad spectroscopic and spatial coverage of DECaLS provides a uniform sample of the local universe, making it particularly effective for identifying low-surface-brightness tidal features and structural distortions indicative of ongoing galaxy-galaxy interactions. [11] [12]

1.5.2 VISTA VIRCam Near-Infrared Photometry

To complement the optical data, near-infrared (NIR) photometry is obtained from the Visible and Infrared Survey Telescope for Astronomy (VISTA), located at the European Southern Observatory's (ESO) Paranal Observatory in Chile. The primary instrument employed is the VISTA Infrared Camera (VIRCam), which utilizes a focal plane array consisting of 16 individual HgCdTe (Mercury-Cadmium-Telluride) detectors. This configuration yields a pixel scale of $0.341''$/pixel and allows for wide-field imaging across the near-infrared spectrum.[13]

2 Sample Selection and Morphological Classification

To achieve a statistically meaningful analysis, it is vital to establish a cleanly classified sample of interacting barred galaxies. The subsequent sections outline the multi-stage filtering and morphological classification methodology applied to our initial parent catalog.

2.1 The Parent Catalog

The sample selection begins with the *Arp-Madore Catalog of Southern Peculiar Galaxies and Associations* [14]. The Arp-Madore catalog is a foundational compilation of structurally anomalous, interacting, and disturbed systems in the southern sky, making it an ideal hunting ground for studying environmentally driven secular evolution. The catalog contains a total of 6,292 unique galaxies exhibiting various degrees of morphological peculiarity, ranging from subtle asymmetric distortions to catastrophic collisional mergers.

Although automated classification pipelines and machine-learning models are highly efficient for processing large-scale uniform sky surveys, they are fundamentally limited when applied to highly irregular or anomalous systems. Because galaxies within the Arp-Madore catalog are inherently peculiar and weirdly shaped, automated algorithms frequently get confused by their complex structural distortions. Such models are prone to producing inaccurate results, either completely misclassifying unique features or overlooking critical, low-surface-brightness details like faint tidal bridges, highly obscured bars, and close projection effects. To overcome these limitations and prevent data contamination, a visual inspection was manually conducted on all 6,292 galaxies in the parent catalog to establish a reliable and physically verified sample.

2.2 Visual Inspection and Classification Criteria

Every galaxy in the parent sample was visually examined utilizing the Legacy Survey Sky Viewer interface, which provides interactive access to deep, multi-band optical imaging from the Dark Energy Camera Legacy Survey (DECaLS) Data Release 9 (DR9) and Data Release 10 (DR10). This comprehensive imaging suite allows for the simultaneous clear identification of bright, small-scale nuclear structures (such as stellar bars) and faint, large-scale structural debris (such as tidal tails or stellar bridges). However, because a subset of the target galaxies fell outside the spatial imaging footprint of the DECaLS DR9 and DR10 surveys, they could not be inspected via the Sky Viewer. For these missing systems, we cross-referenced their coordinates and retrieved their established morphological classifications directly from the NASA/IPAC Extragalactic Database (NED) to ensure complete data coverage across our sample.

A galaxy was classified as "interacting" if it met at least one of the following conditions:

1. The presence of explicit, clear morphological signatures of gravitational interaction, such as a tidal tail, an interstellar bridge connecting to a neighboring system, or highly asymmetric stellar envelopes.

2. The presence of a close, visually confirmed companion galaxy of comparable angular size within an immediate projected radius of $6'$ (6 arcminutes). To rule out line-of-sight chance projections, redshift data were cross-matched and verified utilizing the *Simbad* astronomical database. Systems with matching or tightly clustered redshifts within this angular radius were confirmed as true, physically associated interacting pairs.

Based on these structural and environmental constraints, the 6,292 galaxies were systematically sorted into four distinct analytical subcategories, while the remaining non-compliant galaxies were excluded. The numerical breakdown and operational classification matrix are structured in Table 1.

Table 1: Provisional morphological and environmental breakdown of the initial Arp-Madore parent sample prior to quantitative filtering.

Provisional Candidate Subset	**Initial Visual Environmental Criteria ($r \leq 6'$)**	**Galaxy Count**
Category 1: Candidate Interacting SB	Barred spiral profile; apparent companion or active tidal markers	372
Category 2: Isolated SB	Barred spiral profile; zero apparent companions within search radius	22
Category 3: Group SB	Barred spiral profile; apparent neighbors present but lacking redshifts	732
Category 4: Interacting Spirals	Unbarred spiral profile; apparent physical interaction with companion	173
Excluded Sample	Non-spirals, Ellipticals, S0s, peculiar, compact/dwarfs or mergers	4,993
Total Parent Sample	**Arp-Madore Southern Peculiar Catalog**	**6,292**

2.3 The Four Provisional Morphological Categories

The physical criteria and the respective sample sizes resulting from this initial visual classification are detailed below. These categories represent the baseline candidate pools established during the first-pass optical inspection prior to secondary quantitative filtering.

- **Category 1: Provisional Interacting Barred Spirals (Candidate Interacting SB)** [372 **Galaxies**]
  This group constitutes our primary active candidate pool. These are disk systems that visually appeared to exhibit both a central stellar bar and features of gravitational interaction with an adjacent companion. This pool was selected as the starting point for our detailed downstream analysis to study how tidal forces alter bar dimensions. All systems without obvious tidal distortions were provisionally noted to have a companion within a projected radius of $6'$ based on positional and redshift data from Simbad. However, as detailed in subsequent sections, these 372 galaxies were treated as raw candidates; this sample was later subjected to multi-wavelength data constraints and quantitative profile fitting, which significantly narrowed down the final confirmed sample.

- **Category 2: Isolated Barred Spirals (Isolated SB)** [22 **Galaxies**]
  These galaxies displayed well-defined stellar bars in the initial visual pass but appeared completely isolated, lacking any visible companion within a $6'$ envelope or any signature of historical tidally induced disturbance. While this sample serves as a baseline for internal secular bar growth, these systems were not processed further in the downstream quantitative pipeline, as the primary focus of this study remains strictly bound to active interacting environments.

- **Category 3: Group Barred Spirals with Ambiguous Environment (Group SB)** [732 **Galaxies**]
  This subset contains galaxies with apparent stellar bars that reside within a group environment due to the close proximity of neighboring galaxies inside our $6'$ spatial search window. However, because these systems lack redshift data in Simbad for the surrounding neighbors, it was impossible to distinguish genuine physical groups from line-of-sight projection effects. To protect the primary analysis from environmental contamination, these systems were segregated and set aside after this initial sorting phase.

- **Category 4: Interacting Unbarred Spirals (Interacting Spirals)** [173 **Galaxies**] These systems were classified as unbarred, spiral galaxies undergoing clear, active interactions with a companion confirmed either via visual tidal markers or matching Simbad redshifts within $6'$.

2.3.1 Exclusion Criteria

The remaining 4,993 galaxies from the parent catalog were excluded from further study. Because our research framework is strictly fixed on understanding the mechanics of stellar bars within spiral disks, systems that lacked clear disk components or regular structures were non-compliant.

The excluded pile primarily consists of regular elliptical galaxies (E), lenticular galaxies ($S0$) devoid of open spiral structures, unclassifiable dwarf spheroidals, and mergers. In these mergers, the intense gravitational potential has completely disrupted the host galaxy's disk plane, tearing apart any pre-existing stellar bars or spiral arm structures and turning the system into a highly irregular, unclassifiable object.

Following this initial structural classification, all subsequent analytical work was focused exclusively on the 372 galaxies identified within Category 1. These systems were prioritized as they uniquely satisfied the core environmental and morphological conditions required for our study of bar evolution in interacting systems.

3 Data Refinement and Multi-Wavelength Optimization

3.1 Limitations of Optical r-band Photometry and Isophote Fitting Failure

Initially, quantitative structural analysis and surface photometry were attempted using the r-band optical imaging data obtained from the Dark Energy Camera Legacy Survey (DECaLS). The r-band is highly sensitive to young stellar populations and faint, low-surface-brightness features, making it a standard choice for identifying large-scale tidal features. However, applying automated isophotal ellipse fitting routines to the r-band data of these 372 candidate interacting systems presented severe systematic complications.

First, approximately one-third of the 372 candidate galaxies were altogether missing from the DECaLS footprint, limiting the continuity of the sample. Second, for the galaxies that were successfully imaged, the presence of localized interstellar dust grains in the circumnuclear and disc regions introduced significant differential dust extinction. This high optical depth effectively obscured, patched, or heavily distorted the true distribution of the underlying stellar populations forming the bar structures.

Furthermore, because these systems are actively undergoing gravitational interactions, the optical r-band light profiles are heavily contaminated by young, blue star-forming regions triggered by tidal shocks, alongside prominent asymmetric structures such as tidal tails, bridges, and overlapping debris stellar envelopes. When the isophote fitting algorithm attempted to trace concentric ellipses from the core outward, these highly irregular, non-axisymmetric tidal features heavily distorted the minimization routine. The fitting routine failed to converge stably or produced highly erratic ellipticity and position angle variations that completely masked or fabricated structural signatures, making it impossible to systematically isolate the true stellar bars.

3.2 The Shift to Near-Infrared VISTA K_s-band Imaging

To circumvent the limitations imposed by dust obscuration and tidal starburst noise, the photometric analysis was transitioned to the near-infrared (NIR) K_s-band imaging data from the Visible and Infrared Survey Telescope for Astronomy (VISTA).

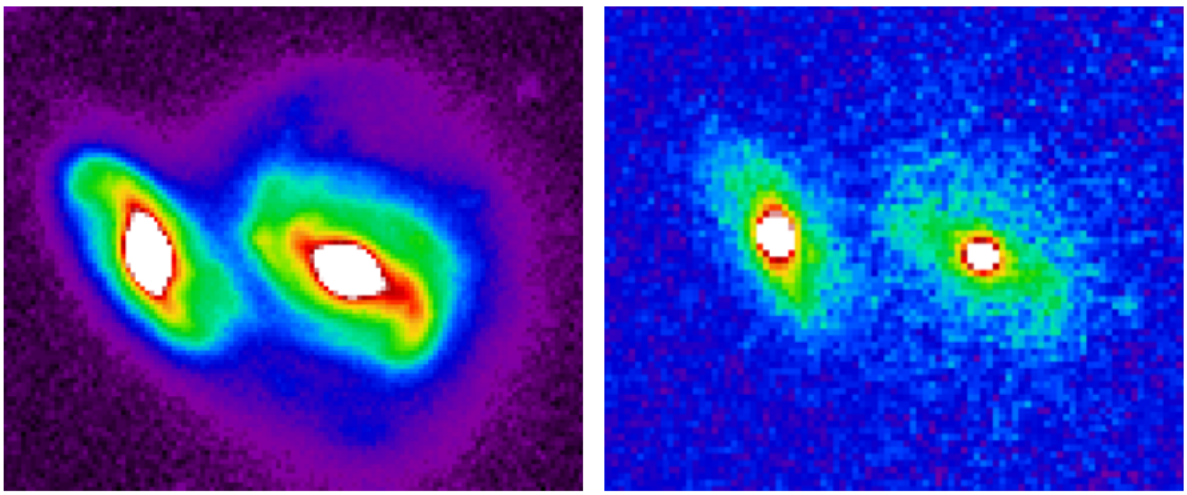


Figure 3: Photometric comparison of an interacting galaxy pair (AM 0011-431) captured in the optical r-band (left) and the near-infrared K_s-band (right), illustrating the physical motivation for utilizing near-infrared datasets for structural parameterization. The optical r-band image is dominated by emission from active star-forming regions, young stellar populations, and localized structural distortions, while being highly susceptible to severe dust obscuration and extinction across the disk and nuclear regions. Conversely, the K_s-band image effectively penetrates the dust lanes, revealing the underlying, unattenuated stellar mass distribution.

The utilization of the NIR K_s-band ($\lambda \approx 2.15\,\mu$m) is physically motivated by two primary factors in galaxy evolution and dynamics:

1. **Minimal Dust Extinction:** The scattering efficiency of interstellar dust grains drops sharply at longer wavelengths. This transparency allows the imaging to penetrate thick circumnuclear dust lanes, providing an unobstructed view of the central galactic structure.

2. **Tracing Old Stellar Populations:** While optical bands are dominated by a small mass fraction of young, short-lived, high-luminosity OB stars tracing localized starbursts, the K_s-band directly traces the light emitted by mature, low-mass K and M giant stars. These stars constitute the bulk of a galaxy's total stellar mass and trace the deep, underlying gravitational potential well of the disk. Consequently, the VISTA K_s-band images reveal the smooth, structural skeleton of the stellar bar and disk, remaining largely unaffected by localized star formation or transient gas dynamics.

3.3 Secondary Filtering and Final Sample Yield

Using the VISTA K_s-band dataset, a secondary, visual inspection was executed exclusively on the 372 candidate systems from Category 1. Galaxies that exhibited completely unbarred profiles or unresolvable nuclear regions under the near-infrared regime were excluded. This stage effectively narrowed the sample down from 372 candidates to 150 structurally promising systems.

Quantitative isophotal ellipse fitting using Photutils was subsequently executed on these 150 systems. The resulting structural profiles, specifically tracking the radial variations of ellipticity (ϵ) and position angle (PA) were examined to isolate a definitive bar signature. In a standard barred galaxy, a genuine bar manifests as a synchronized, sharp rise in ellipticity that maintains a stable, nearly constant position angle across the bar's spatial extent, followed by a sharp drop in ellipticity and a sudden position angle drift at the transition boundary to the disk or spiral arms.

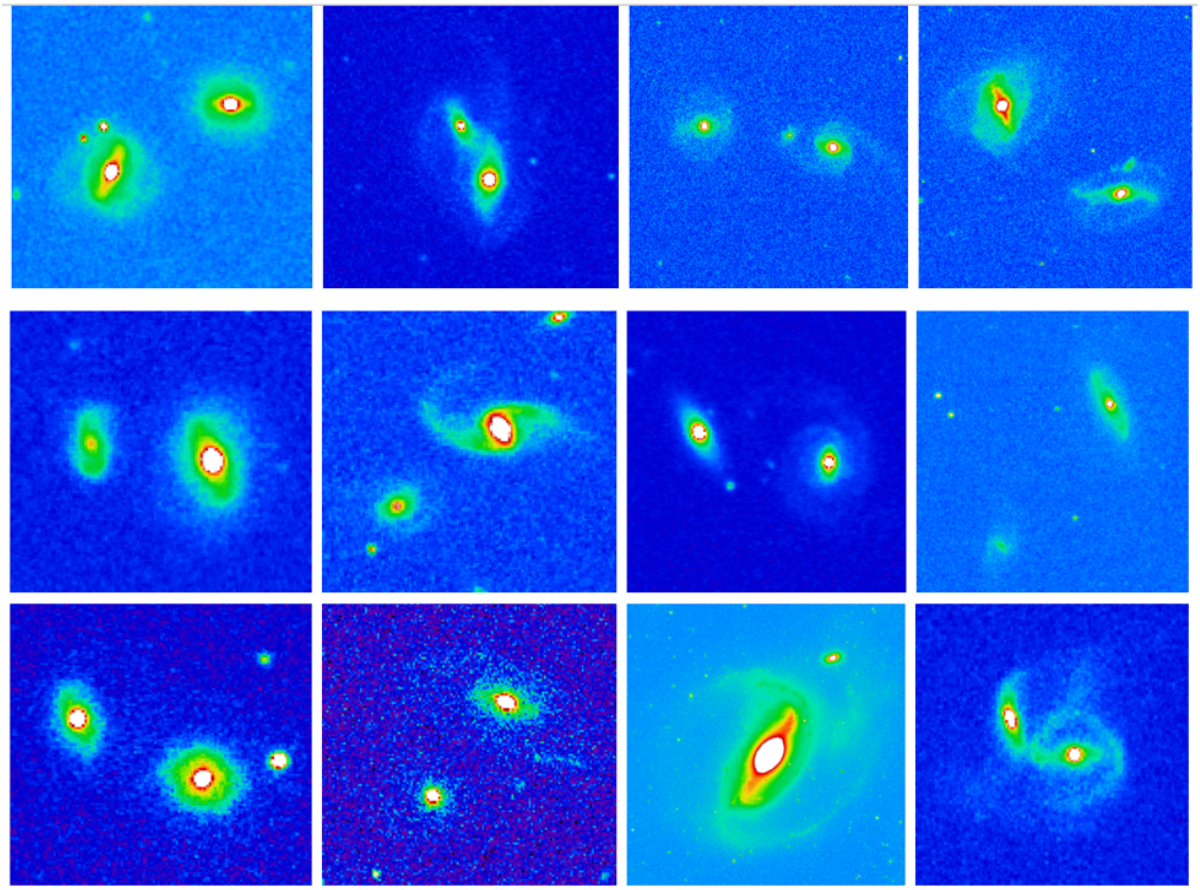


Figure 4: K_s-band images of the final sample galaxy systems used in our analysis. The top row shows pairs where both galaxies have bars. The middle and bottom rows show pairs where only one galaxy has a bar.

Galaxies whose profiles failed to fulfill these strict geometric criteria were excluded from the analysis. This multi-wavelength quantitative filtering resulted in a final verified sample of 75 interacting systems, which break down into two distinct environmental configurations:

- **Single-Barred Systems (70 Systems):** Interacting pairs in which only one host galaxy possesses a quantitatively confirmed stellar bar structure while its companion remains unbarred.
- **Dual-Barred Systems (5 Systems):** Co-interacting pairs in which both the primary galaxy and its immediate companion harbor distinct, confirmed stellar bars.

Accounting for both components of the double-barred pairs, this workflow yielded a total of 80 individual barred galaxies.

4 Ellipse Fitting and Isophotal Analysis

4.1 Overview of the Photutils Ellipse Fitting Tool

To extract quantitative structural parameters of the stellar bars, we utilized the `photutils.isophote` package [15], a specialized Python library developed by the Space Telescope Science Institute (STScI) for astronomical photometry. The software functions by analyzing the two-dimensional light distribution of a galaxy and fitting a series of concentric ellipses that match the contours of equal light intensity, known as isophotes.

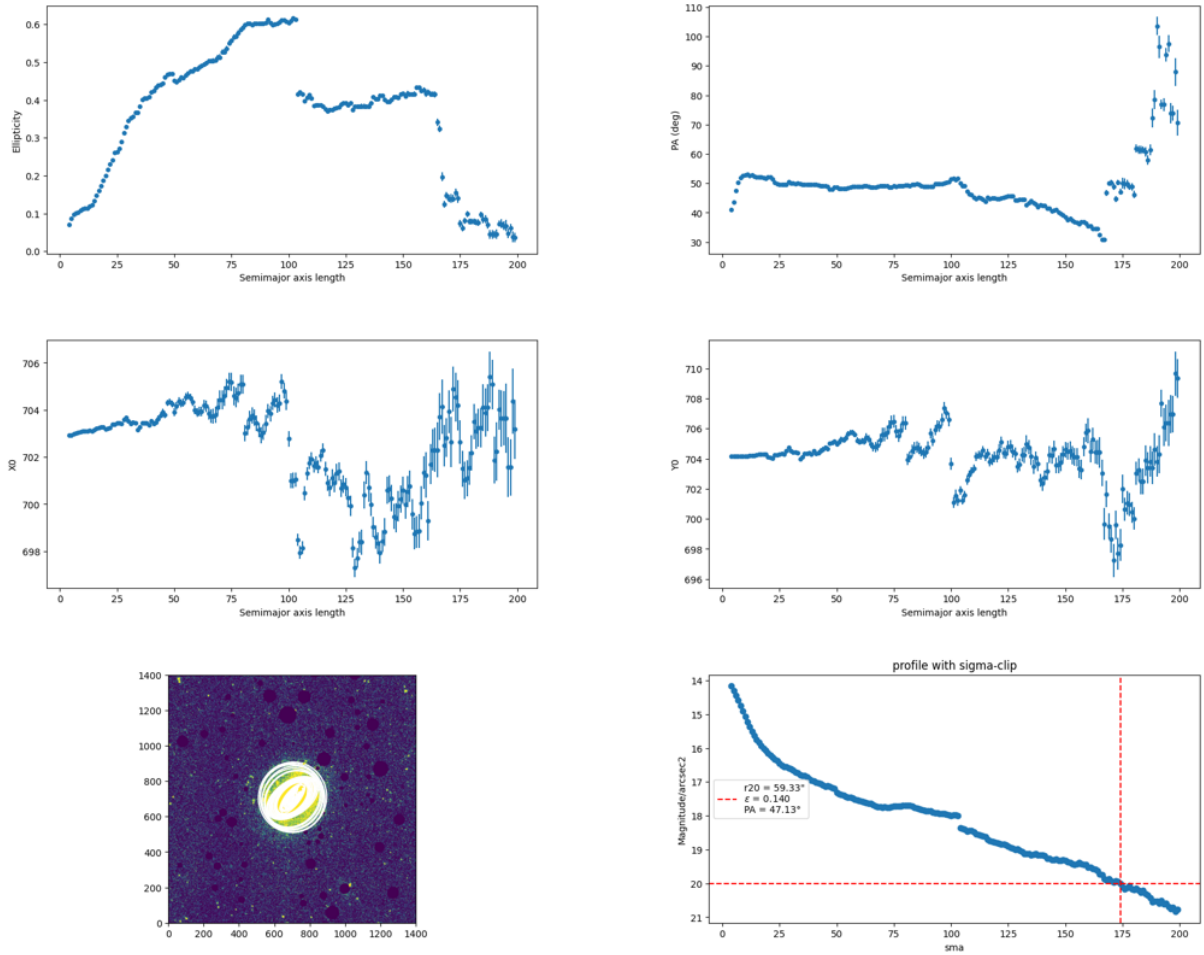


Figure 5: Isophotal analysis and ellipse-fitting profiles generated using `photutils` for the galaxy AM 0055-491. The multi-panel plot displays structural parameters as a function of the semi-major axis length (SMA) in pixels.
Top row: Ellipticity vs. SMA (left) and Position Angle (PA in degrees) vs. SMA (right).
Middle row: Center variations of the fitted ellipses shown using the X_0 (x center) vs. SMA (left) and Y_0 (y center) vs. SMA (right).
Bottom row: Best-fit elliptical isophotes overlaid on the galaxy's intensity image (left) and the surface brightness profile in mag/arcsec^2 vs. SMA (right)

The algorithm operates iteratively, starting from user-defined initial guesses for the galaxy's center, ellipticity, and position angle. As it moves radially outward, it samples the pixel intensities along the path of a trial ellipse. The software then calculates how much the intensity fluctuates along that path and uses those variations to adjust the shape, centering, and orientation of the next ellipse. This process is repeated until it finds the optimal geometry that best traces the stellar distribution at that specific radius. By doing this across the entire face of the galaxy, the software generates a continuous profile showing how the galaxy's shape changes from its nuclear core out to its low-surface-brightness outskirts. [16]

4.2 Masking Star Contamination and Companion Isolation

Because the ellipse fitting routine operates by evaluating pixel intensities along a geometric path, any extraneous source of light that is not part of the target galaxy's smooth stellar body will severely bias the fit. If the algorithm encounters a bright, unmasked object, the ellipse will artificially bulge or shift its center toward that light source, producing false geometric features.

To ensure clean structural measurements, a masking pipeline was implemented before running the sky background subtraction:

- **Foreground Stars and Background Galaxies:** Line-of-sight foreground stars from our own Milky Way and random background galaxies frequently overlap with the target galaxy's disk. These point sources and distinct light profiles were identified and covered with a binary bad-pixel mask, instructing the fitting algorithm to completely ignore those pixels during its intensity evaluations.
- **Isolating the Companion Galaxy:** In an interacting system, the close proximity of the companion galaxy presents a massive challenge for standard photometry. As the fitting ellipses expand outward to measure the outer disk of the primary galaxy, they inevitably begin to overlap with the light profile of the incoming companion. Without intervention, the algorithm will attempt to stretch the primary galaxy's outer isophotes to encompass the companion, resulting in a failure of the fit. To isolate the primary system, the companion galaxy was entirely masked. This forced the software to compute the optimal ellipses based strictly on the unperturbed halves and clean regions of the primary galaxy's disk.

4.3 Sky Background Subtraction

Before running the ellipse fitting pipeline, a precise sky background subtraction must be executed on every image. Pixels in a raw astronomical image do not just record the light from the target galaxy; they also capture diffuse ambient light from the night sky, atmospheric glow, and faint unresolved cosmic backgrounds.

If this global background light is not accurately measured and subtracted, the outer, fainter regions of the galaxy become highly corrupted. Because the surface brightness of a galaxy drops off drastically at large radii, the faint light from the outer disk quickly becomes comparable to the brightness of the sky background itself. If a residual sky gradient or a positive background offset is left in the image, the fitting algorithm will get confused. Instead of fitting the true circular or elliptical contours of the outer disk, the ellipses will violently stretch out, warp, or align themselves with the background noise. This completely ruins the measurements of the disk's true orientation and outer radius. To prevent this, the local background level was carefully estimated from empty sky regions surrounding the targets and subtracted from the science frames.

Through this combination of clean background subtraction and strict masking, the fitting tool was able to reliably isolate the true structural features of the primary galaxy, providing clean, uncontaminated profiles of ellipticity and position angle across the bar region. [17]

4.4 Galactic Parameters

The direct output of the isophotal fitting sequence provides the mean intensity along each fitted ellipse as a function of its semi-major axis (SMA). To convert these raw pixel intensity values into physically meaningful astronomical units, the intensity along each isophote was converted into surface brightness (μ_K) expressed in units of magnitudes per square arcsecond (mag/arcsec^2). This calibration was performed using the standard photometric relation:

$$\mu_K = -2.5 \log_{10} \left(\frac{\text{counts}}{\text{pixel scale}^2} \right) + Z_p \tag{1}$$

where *counts* represents the mean pixel intensity along a given isophote, Z_p is the instrumental photometric zero-point assigned to the VISTA K_s-band science frames, and the *pixel scale* for the VISTA instrument configuration is taken as $0.341''$/pixel. [18] [19]

By plotting this calibrated surface brightness against the semi-major axis, we generated continuous radial surface brightness profiles for each system. These profiles were used to determine the R_{20} radius of the host galaxies. The R_{20} radius is defined as the limiting isophotal radius where the galactic surface brightness drops to exactly $\mu_K = 20\,\text{mag/arcsec}^2$. This metric serves as a reliable proxy for the total physical size of the galactic disk in near-infrared imaging.

Once the R_{20} boundary was determined, the properties of the ellipse matching this specific radius was used to lock in the host galaxy's structural orientation parameters: [20]

1. **Galaxy Position Angle ($\mathbf{PA_{gal}}$):** Taken directly from the orientation of the major axis of the outer disk isophotes at R_{20}, representing the true line-of-nodes of the galactic disk on the sky plane.

2. **Galaxy Inclination (i):** Calculated analytically from the outer disk ellipticity (ϵ_{disk}) measured at the R_{20} boundary, assuming an intrinsically thin circular disk viewed at an angle:
$$\cos(i) = 1 - \epsilon_{\text{disk}} \quad [1] \tag{2}$$

4.5 Bar Deprojection

Simultaneously, the behavior of ellipticity (ϵ) and position angle (PA) as a function of SMA was examined at smaller radii to isolate the properties of the central bar structure. The raw visual spatial extent of the bar in these plots provides the observed bar length (L_{obs}) and the observed bar position angle ($\text{PA}_{\text{bar, obs}}$).

However, because galaxies in the universe are randomly oriented relative to our line of sight, they are rarely viewed perfectly face-on ($i = 0^\circ$). When a tilted galactic disk containing a linear stellar bar is projected onto the two-dimensional plane of the sky, projection effects severely distort its apparent geometry. The observed bar undergoes foreshortening, making it appear artificially shorter than its true physical length. Evaluating raw, projected bar lengths across a sample with varying inclinations would introduce severe systematic biases, invalidating any statistical comparisons.

To recover the true, physical dimensions of the stellar structures, the observed parameters must be geometrically projected back onto the intrinsic, face-on plane of the host galaxy's disk. Using the galaxy inclination (i) and defining α as the angular difference between the galaxy's outer disk position angle and the observed bar position angle ($\alpha = |\text{PA}_{\text{gal}} - \text{PA}_{\text{bar, obs}}|$), the true, deprojected physical bar length (L_{bar}) was calculated using the following coordinate transformation:

$$L_{\text{bar}} = L_{\text{obs}} \sqrt{\sin^2(\alpha) \sec^2(i) + \cos^2(\alpha)} \quad [21] \tag{3}$$

This deprojected bar length (L_{bar}) represents the true physical scale of the bar structure. It was subsequently normalized against the outer disk radius (R_{20}) to compute the scale-independent, normalized bar length ($L_{\text{norm}} = L_{\text{bar}}/R_{20}$), enabling a direct comparison of bar strength across galaxies of vastly different physical sizes.

4.6 Sample Refinement

The initial observational sample consisted of 75 interacting systems, partitioned into 70 single-barred systems and 5 dual-barred systems. To ensure the physical validity of the bar deprojection, an inclination filter was applied. Systems featuring galaxies with galactic disk inclinations outside the range of $20^\circ \leq i \leq 60^\circ$ were excluded from the final analysis, as any targets outside this boundary cannot be reliably deprojected.

The justification for this deprojection limit is twofold:

1. **Upper Bound Restriction ($i > 60^\circ$):** In highly inclined or edge-on systems, the projection of the disk along the line of sight, coupled with vertical thickness effects and heavy dust lane obscuration, prevents a reliable deprojection of the bar's true physical length. [22]

2. **Lower Bound Restriction ($i < 20^\circ$):** For nearly face-on systems, the observed disk is highly circular. This makes determining the disk's position angle (PA_{disk}) and exact inclination angle mathematically unstable. Because deprojection formulas rely heavily on these angles, even minor measurement errors are highly amplified, rendering the resulting deprojected bar parameters unreliable. [23]

Following the application of this inclination threshold, the final dataset yielded 62 unique data points across 57 interacting systems, comprising 52 single-barred systems (52 galaxies) and 5 dual-barred systems (10 galaxies).

5 Stellar Mass Derivation

5.1 Photometric Systems: Vega and AB

To calculate the total stellar mass of the galaxies, the total flux inside the R_{20} boundary must first be converted into an absolute physical luminosity. This process requires working across two distinct astronomical magnitude scales: the Vega system and the AB (Absolute) magnitude system.

The Vega system is an operational, instrument-based scale where the bright A0V star Vega is defined as the baseline zero-point ($m = 0$) across all photometric filters. Because historical near-infrared datasets, including the VISTA K_s-band imagery, are traditionally calibrated using stars, their raw apparent magnitudes are natively anchored to the Vega system. [24] Conversely, the AB magnitude system is a monochromatic flux scale defined such that a flux density of 3631 Jy corresponds to a magnitude of 0 at all frequencies. The AB system is strictly decoupled from stellar models, making it the standard choice for calculating absolute physical properties like solar luminosities ($L/L_\odot$). [25]

5.2 Luminosity and Distance Calculations

The first step in mass derivation involved summing the total pixel intensities (total counts) enclosed within the defined R_{20} boundary for both the primary and companion galaxies across all 75 interacting systems. These integrated counts were converted into an integrated apparent K_s-band magnitude (m_K) via the calibrated zero-point relation (eqn. 1).

Because the apparent magnitude m_K is natively in the Vega system, the corresponding absolute magnitude (M_K) was evaluated using the standard distance modulus relation:

$$M_K = m_K - 5\log_{10}(D) + 5 \quad [18] \tag{4}$$

where the luminosity distance D is expressed in units of parsecs (pc). To compute D for each system, the cosmological redshift (z) was extracted from the Simbad database. Assuming a standard Hubble constant of $H_0 = 70\,\mathrm{km/s/Mpc}$, the distance was calculated analytically via Hubble's Law:

$$D = \frac{z \cdot c}{H_0} \times 10^6 \quad [26] \tag{5}$$

where c represents the speed of light in vacuum ($\approx 3 \times 10^5\,\mathrm{km/s}$), and the factor of 10^6 converts the resulting distance from megaparsecs (Mpc) to parsecs (pc).

To transition from the native instrument scale to a physical scale, the absolute Vega magnitude (M_K) was converted to the AB system (M_{AB}) by adding a standard offset characteristic of the K-band filter profile:

$$M_{\mathrm{AB}} = M_K + 1.85 \tag{6}$$

Using the absolute magnitude in the AB system, the total galactic luminosity (L) expressed in units of solar luminosity was determined using the following relation:

$$\log_{10}\left(\frac{L}{L_\odot}\right) = -0.4 \cdot (M_{\mathrm{AB}} - M_\odot) \tag{7}$$

where $L_\odot$ represents the solar luminosity, and $M_\odot$ denotes the absolute magnitude of the Sun in the K_s-band under the AB system. For the purpose of this calculation, the value of $M_\odot$ was taken to be 3.27. [27]

5.3 Mass Calculations and Ratios

Finally, the physical stellar mass ($M_\star$) of each galaxy was derived by applying a standard near-infrared mass-to-light ratio ($\Upsilon_\star^K = 0.6$):

$$M_\star = 0.6 \times L \quad [28] \tag{8}$$

The near-infrared K_s-band is exceptionally stable for mass derivations because its mass-to-light ratio remains largely uniform across different galaxy types, independent of active starbursts or varying stellar ages.

This computation was applied uniformly to all galaxies across the 62 verified systems, explicitly capturing the individual masses of both the primary host and its interacting companion. To quantify the dynamic scale of the gravitational interaction, the mass ratio (q) of each interacting pair was computed using the standard convention:

$$q = \frac{M_{\mathrm{lighter}}}{M_{\mathrm{heavier}}} \tag{9}$$

By structuring the ratio such that the lighter component sits in the numerator, the resulting value of q is strictly bounded between 0 and 1, providing a standardized baseline to classify interactions into major ($q \geq 0.3$) or minor mergers ($q < 0.3$). [29]

6 Results and Discussions

6.1 Distribution of Normalized Bar Lengths

To establish the baseline structural properties of the interacting galaxy sample, we first analyzed the frequency distribution of the normalized bar lengths, L_{bar}/R_{20}. The global sample exhibits

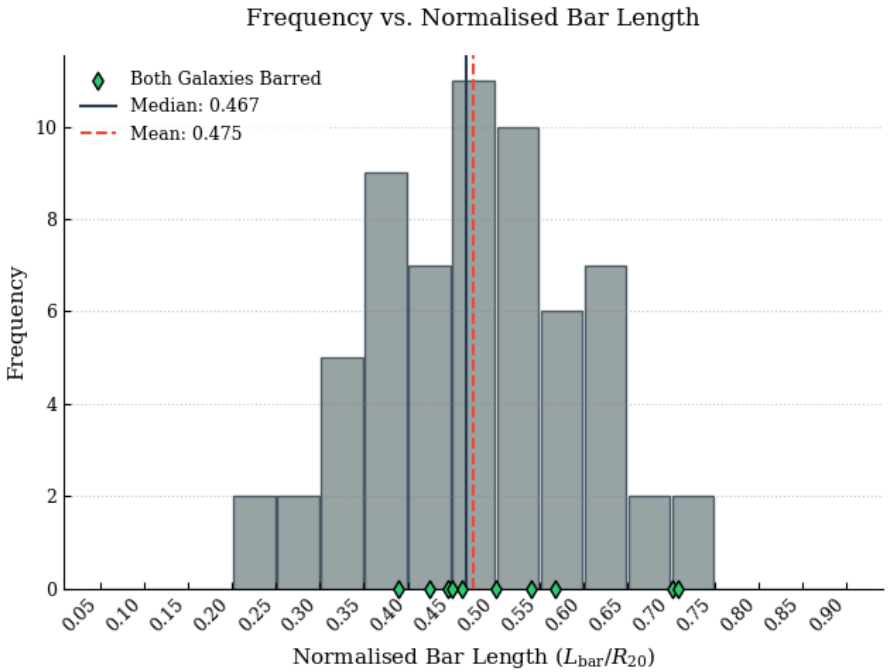


Figure 6: Frequncy vs. Normalised Bar Length Histogram

a well-defined, single-peaked distribution centered around a median normalized length of 0.467 and a mean of 0.475.

Physically, a median value of approximately ~ 0.47 indicates that, on average, stellar bars in interacting systems extend to nearly half the radius of the host galaxy's optical disk (as defined by the R_{20} isophotal radius). This substantial length suggests that tidal interactions are highly efficient at extracting angular momentum from the stellar and gaseous disk. This momentum loss drives strong global instabilities, reinforcing the elongated orbits that support the bar structure.

6.2 Galaxy Mass Ratio

We evaluated the mass ratios (q) of the interacting pairs to determine how the relative gravitational influence of the companion dictates bar formation.

For systems where only a single galaxy is barred, the distribution gives a median value of $q = 0.36$. Conversely, for dual-barred systems, the distribution shifts toward major-merger symmetry. The individual mass ratios measured for these five dual-barred pairs are $0.36, 0.44, 0.595, 0.788$, and 0.89, yielding a sample median of 0.595. Every single dual-barred system lies on or above the median mass ratio of the single-barred population.

By adopting the standard physical threshold where a mass ratio of $q \geq 0.3$ constitutes a major merger, we can categorize the 62 galaxies in our sample to isolate the environmental dependence of bar triggering:

- **Major Mergers ($q \geq 0.3$):** 37 galaxy systems in total. Within this category, 5 galaxy systems belong to dual-barred systems, yielding an empirical probability of $P(\text{Dual Bar} \mid \text{Major Merger}) \approx 13.5\%$ (10/42).
- **Minor Mergers ($q < 0.3$):** 20 galaxies in total. 0% of the galaxies in this category belongs to dual-barred systems.

This clear bifurcation reveals a physical constraint on bar co-evolution. Minor mergers ($q < 0.3$) lack the symmetric tidal forces required to simultaneously trigger or stabilize bars in

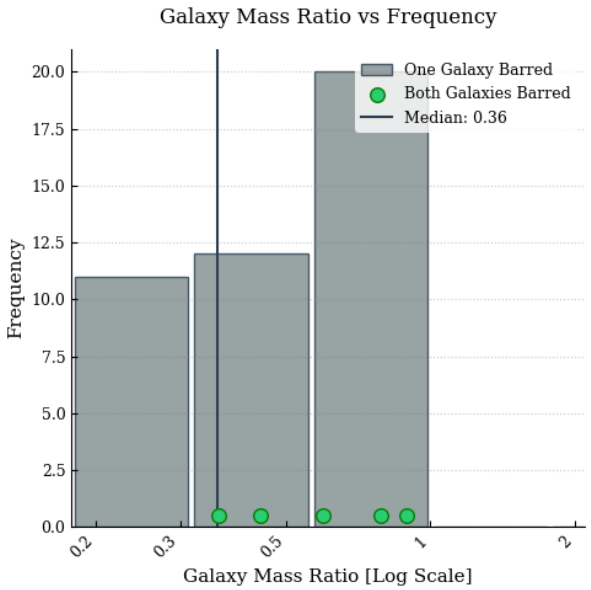


Figure 7: Frequncy vs. Galaxy Mass Ratio Histogram

both components. In an asymmetric potential, the massive primary galaxy easily absorbs the weak perturbation of the satellite, whereas the satellite itself experiences violent tidal stripping that disrupts the disk coherence necessary to sustain a stable bar. Dual bars, therefore, require a mutual and relatively balanced gravitational potential found exclusively in major-merger environments.

6.3 Correlation Between Bar Length and Mass Ratio

To test how interaction strength directly influences bar growth, we plotted the normalized bar length (L_{bar}/R_{20}) against the log-scale galaxy mass ratio (q). This revealed two distinct, opposing dynamical behaviors between the two populations.

6.3.1 Single-Barred Systems (Positive Trend)

The single-barred galaxies exhibit a weak positive correlation, characterized by the following parameters:

$$r = 0.28, \quad \rho = 0.27, \quad m = 0.86$$

While the correlation coefficients indicate a statistically weak linear relationship ($r \approx 0.3$), the positive slope ($m = 0.86$) is physically expected. As the mass ratio increases toward unity ($q \to 1.0$), the tidal forces exerted on the barred galaxy grows stronger. This heightened forces amplify non-circular stellar motion in the disk, feeding and reinforcing the elongated x_1 orbits that constitute the bar, thereby extending its normalized physical length.

6.3.2 Dual-Barred Systems (Negative Trend)

In contrast, the dual-barred systems display a negative correlation:

$$r = -0.29, \quad \rho = -0.07, \quad m = -0.40$$

This is a counter-intuitive but physically revealing result, indicating that as the mass ratio approaches unity ($q \to 1.0$) in dual-barred systems, the bars actually tend to be shorter.

In near-equal mass major mergers, the mutual tidal interactions become exceedingly violent. Rather than gently torquing the disk, these near-symmetric encounters drive massive, rapid gas

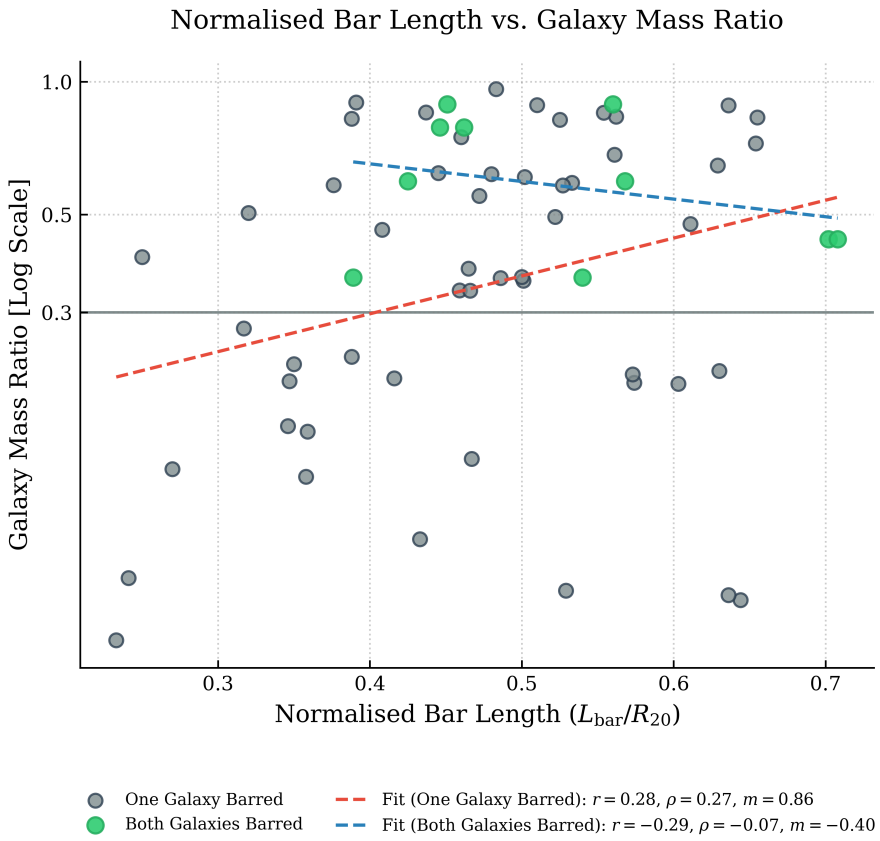


Figure 8: Normalised Bar Length vs. Galaxy Mass Ratio scatter plot

inflows toward the central regions of both galaxies, forming a dense Central Mass Concentration (CMC). A sufficiently massive CMC alters the central gravitational potential, scattering stars out of their bar-supporting x_1 orbits and onto chaotic or circular orbits. This orbital heating can cause the bar to buckle, weaken, or physically shorten compared to more asymmetric major mergers.

It is also important to address the statistical limitations of this trend. With a Spearman rank coefficient of $\rho = -0.07$ and a small sample size of $N = 10$ for the dual-barred galaxies, the negative trend lacks the statistical significance required to claim a definitive scaling law.

7 Conclusion

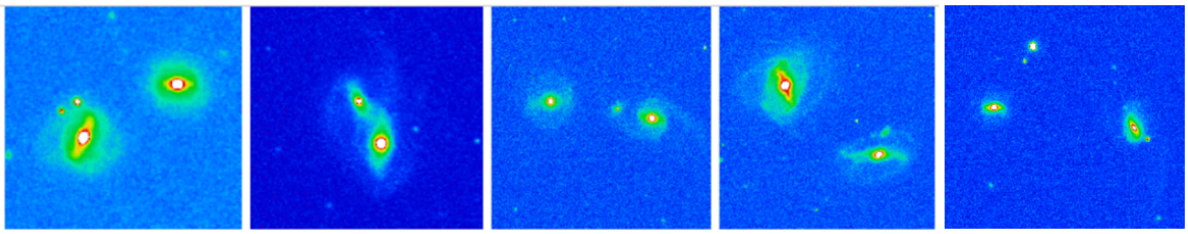


Figure 9: K_s-band images of the dual-barred systems

In this project, we investigated how tidal interactions in galaxy pairs affect the structure and evolution of stellar bars. By using VISTA K_s-band near-infrared data, we managed to look past dust interference and measure the structural properties of 62 galaxies across 57 interacting systems.

Our main finding is that **dual-barred systems, where both galaxies in the pair have a confirmed bar, occur exclusively in major mergers ($q \geq 0.3$).** We did not find a single dual-barred configuration in minor mergers ($q < 0.3$). This strongly suggests that a near-equal mass ratio is necessary to provide the symmetric gravitational conditions needed to trigger and maintain bars in both galaxies simultaneously, without the larger galaxy tidally disrupting the smaller companion.

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